DS 501 · INTRODUCTION TO DATA SCIENCE

Week 05 · Data Cleaning, Wrangling and EDA

Reading Summaries

Two slides per paper: the problem and key ideas, then what to take away and how it connects to this week. Use them after your own first pass, not instead of it.

THE PAPERS

R1 **Kandel et al. 2011**
Required · Wrangler: Data Cleaning by Example and Preview

R2 **Rubin 1976**
Required · When Can We Ignore Why Data Are Missing

01 **Kim et al. 2003**
Optional · A Taxonomy of Dirty Data

02 **Tukey 1977**
Optional · Look at the Data First: Stem-and-Leaf Displays

International Balkan University

Required

Wrangler: Data Cleaning by Example and Preview

Kandel, S., Paepcke, A., Hellerstein, J., & Heer, J. (2011). Wrangler: Interactive visual specification of data transformation scripts. In Proceedings of CHI '11 (pp. 3363–3372). ACM. https://doi.org/10.1145/1978942.1979444

THE PROBLEM

Analysts spend much of their time reshaping data and fixing missing values, typos and odd formats. These steps are hard to specify (e.g. regular expressions), and manual edits in tools like Excel leave no reusable record.

KEY IDEAS

01 Select data in a table; the system suggests a ranked list of possible transforms

02 Each suggestion has a plain-language description and an in-place visual preview before you apply it

03 Every step is saved as a script that can be edited, annotated, exported as code and rerun on new data

Key term: Data wrangling — Reformatting data values or layout, fixing wrong or missing values, and combining sources so analysis can begin

Required

Cleaning Should Produce a Script, Not Just Clean Data

WHAT TO TAKE AWAY

1. In a 12-person study, median task time in Wrangler was over twice as fast as in Excel on all three tasks
2. Users rated previews and suggestions as more useful than editing transform parameters directly
3. Users without a mental model of the transform (e.g. reshaping) got stuck in both tools

THIS WEEK

Wrangler's operators (split, extract, fill, delete, fold/unfold) match pandas steps in 5-2. In the Titanic lab, keep cleaning in a notebook script, not hand edits, so it can be rerun.

LIMITATION

Small study: 12 users, tiny tables (at most 30 rows, 4 columns), three tasks. Gains on large, real data were expected, not measured.

For your half-page note: Why do the authors say the output of wrangling is a script, not only clean data? How does this apply to your Titanic notebook?

Required

When Can We Ignore Why Data Are Missing

Rubin, D. B. (1976). Inference and missing data. Biometrika, 63(3), 581–592. https://doi.org/10.1093/biomet/63.3.581

THE PROBLEM

Analysts often analyse only the values they have and ignore the process that made other values missing. Rubin asks when this is valid, and shows that the answer depends on the type of inference used.

KEY IDEAS

01 Treat missingness as random data: an indicator per value with its own probability model

02 Missing at random (MAR): missingness may depend on observed values, not on the missing ones

03 Observed at random (OAR) adds that missingness does not depend on the observed values either

Key term: Ignorable missingness — When the missingness process can be left out of the model and inference from the observed data is still valid

WEEK 5 · REQUIRED READING 2 · RUBIN 1976

Required

Ignoring Missingness Is an Assumption, Not a Default

WHAT TO TAKE AWAY

1. For likelihood and Bayesian inference, MAR plus distinct parameters lets us ignore the missingness process
2. For sampling-based (frequentist) inference, the data must be both MAR and observed at random
3. These ideas grew into the MCAR, MAR and MNAR labels used in later missing-data work

THIS WEEK

In 5-1 and the Titanic lab, ask why Age or Cabin is missing before you drop rows or fill values. dropna() and fillna() each assume a missingness mechanism.

LIMITATION

Mathematical and abstract. MAR cannot be tested from the observed data alone, so in practice it remains an assumption to argue for.

For your half-page note: For Titanic Age, is MCAR, MAR or MNAR most plausible? What would that mean for dropping rows versus filling the median?

Optional

A Taxonomy of Dirty Data

Kim, W., Choi, B.-J., Hong, E.-K., Kim, S.-K., & Lee, D. (2003). A taxonomy of dirty data. Data Mining and Knowledge Discovery, 7(1), 81–99. https://doi.org/10.1023/A:1021564703268

THE PROBLEM

Data mining and data warehouses depend on clean data, but there was no shared, complete list of the kinds of dirty data. Without one, it is hard to judge what a cleaning tool covers or misses.

KEY IDEAS

01 Classify dirty data in a hierarchy instead of a loose list of examples

02 Top split: missing data, wrong data, and data that is not wrong but still unusable

03 Link each type to how it can be prevented, detected or fixed, e.g. by database checks

Key term: Dirty data — Data that is missing, wrong, or correct but unusable as stored, e.g. inconsistent formats across sources

Optional

Not All Dirty Data Is Missing or Wrong

WHAT TO TAKE AWAY

1. Missing values are only one kind of dirty data; wrong values and unusable values need other methods
2. Some wrong data can be blocked by database constraints; much of it cannot and needs later checks
3. Correct values can still be unusable, e.g. different codes or spellings for the same thing

THIS WEEK

Use the taxonomy as a checklist for 5-1 and the lab: find missing values in messy Titanic, then wrong ones, then valid but inconsistent ones (e.g. mixed spellings).

LIMITATION

Built from a database and data-warehouse view; it lists problem types but gives no measured evidence of how often each occurs.

To think about: Give one Titanic example for each top-level class: missing, wrong, and not wrong but unusable. How would you detect each in pandas?

Optional

Look at the Data First: Stem-and-Leaf Displays

Tukey, J. W. (1977). Exploratory data analysis (Chapter 1: Scratching down numbers). Addison-Wesley.

THE PROBLEM

Analysts often jump to formal tests and models without first looking at the data. Lists of raw numbers hide the shape of a distribution, gaps and odd values.

KEY IDEAS

01 Exploratory analysis is detective work: look for clues before you confirm anything

02 Exploratory and confirmatory analysis are both needed; exploration comes first

03 Stem-and-leaf display: split numbers into stem and leaf to show shape and keep the values

Key term: Stem-and-leaf display — A text chart: leading digits form the stems, the next digit of each value is written as a leaf beside its stem

Optional

A Quick Display Can Show What a Table of Numbers Hides

WHAT TO TAKE AWAY

1. A stem-and-leaf shows a distribution's shape like a histogram while keeping the actual values
2. It makes clusters, gaps and unusual values easy to spot, and can be drawn by hand
3. Simple displays help decide which questions are worth testing formally later

THIS WEEK

This is the mindset of lesson 5-3. Before modelling survival, look at Titanic Age and Fare with describe(), histograms or a stem-and-leaf, and note gaps and odd values.

LIMITATION

Written for pencil-and-paper work on small data; stem-and-leaf displays become impractical for very large datasets.

To think about: Draw a stem-and-leaf of 20 Titanic ages. What does it show that the mean and standard deviation do not?

Writing Your Half-Page Note

01

The problem

What was broken or missing before this paper? One or two sentences.

02

The key idea

The one idea you would keep if you forgot everything else. Your own words.

03

The connection

Where it shows up this week: a lesson, the lab, or your project data.

Half a page means about 150–200 words. Bring it to the next class; one quiz question may refer to the required reading.